

Introduction

We propose a deep learning-based framework for quality assessment of physiological signals, by analyzing signals via smaller segments

Background

Physiological signals are collected concurrently during functional MRI scans to support a comprehensive understanding of the subject's physical state, but the utility of these signals depends on their quality.

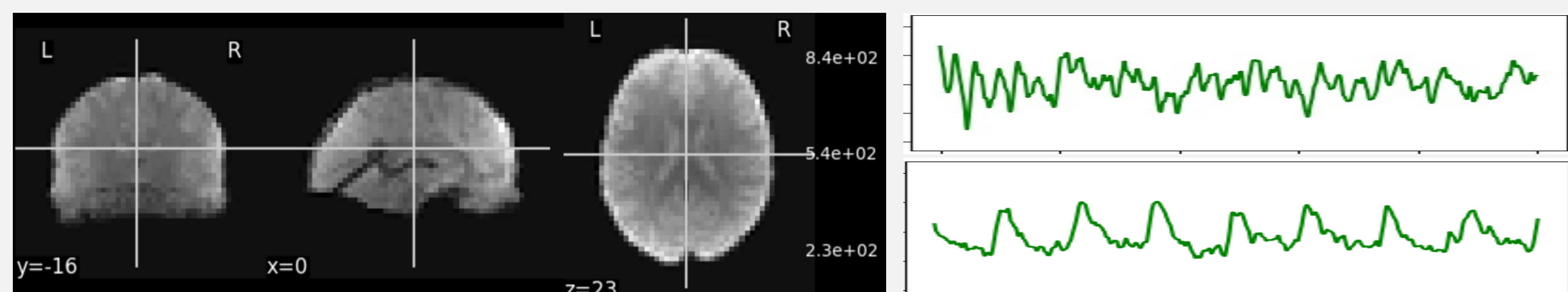


Figure 1: Examples of fMRI scan image with physiological signals

Motivation

Motivation 1: Existing manual quality assessment of physiological data is expensive, time-consuming and involves subjectivity → Can we automate this process with machine learning models?

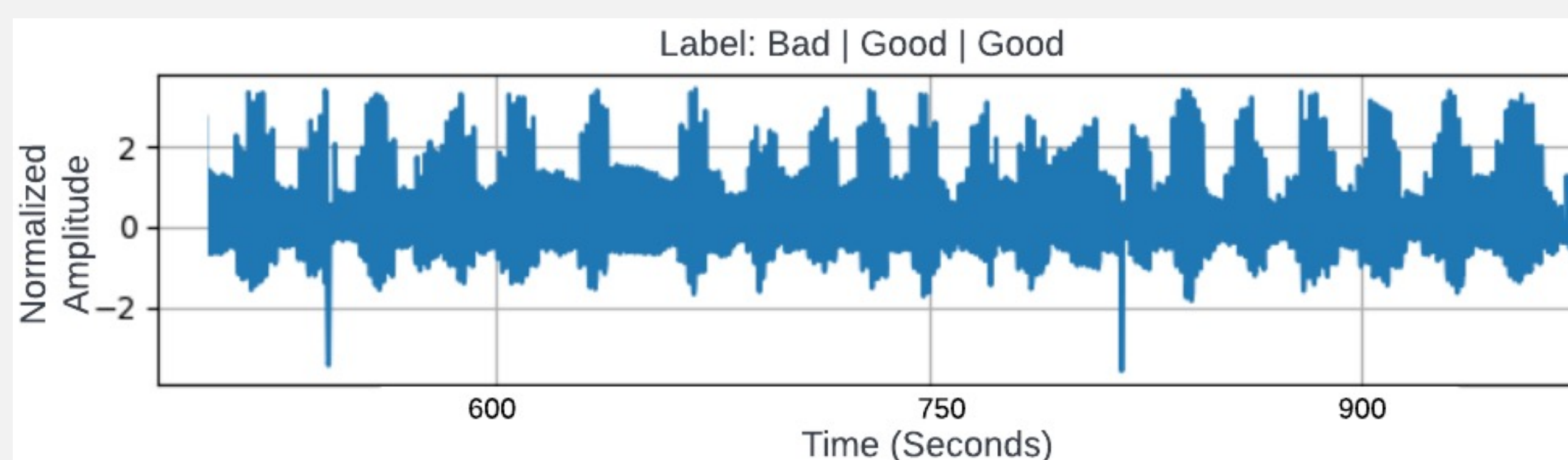


Figure 2: Example segment of a cardiac signal with differing manual annotations from three experts

Preliminary Analysis: Why We Can't Operate in Label-Free Settings

- Unsupervised methods (spectral clustering, K-means) partially separate bad signals after dimensionality reduction (t-SNE, PCA).
- Complex physiological artifacts limit this approach, motivating label-aware learning.

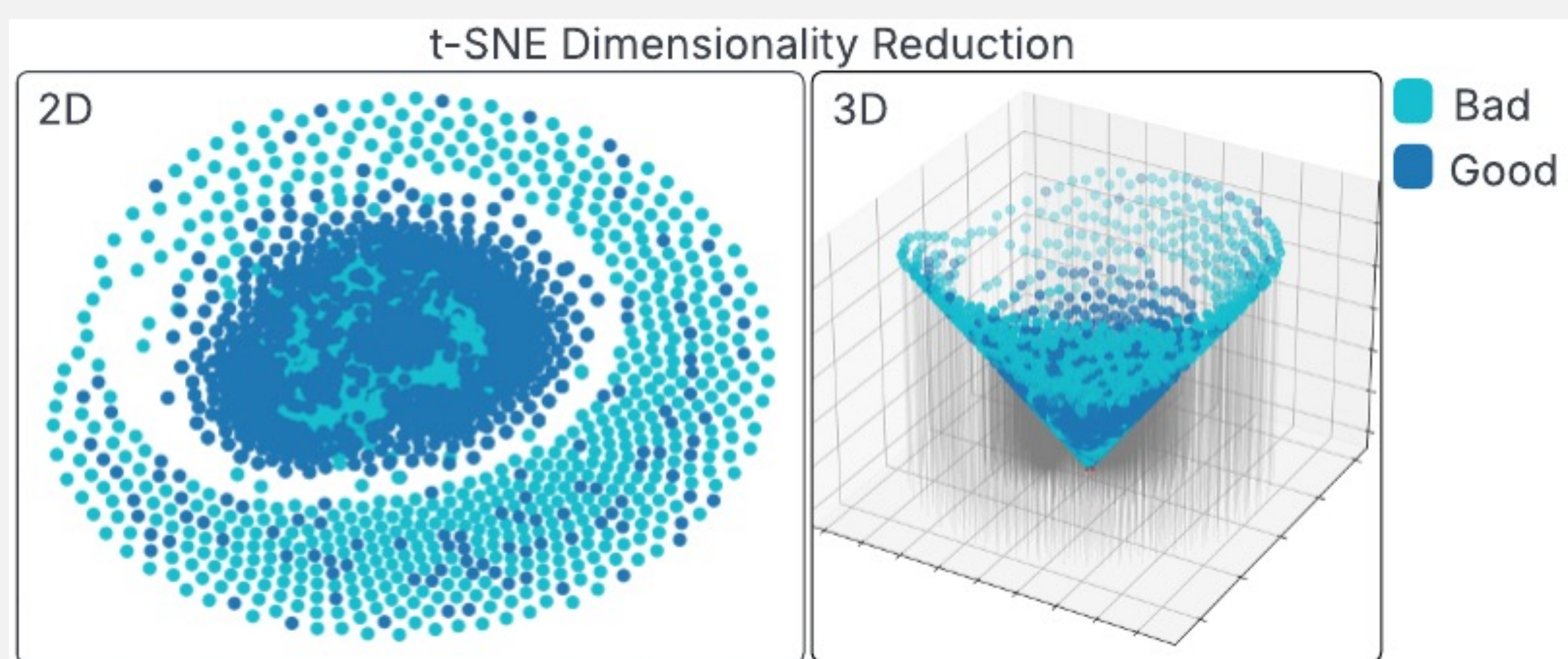


Figure 3: True distribution with T-SNE dimensionality reduction to 2D and 3D

Motivation 2: Manual annotation follows a segment-by-segment analysis of the full signal → Can we design models that replicate this approach with informative segment-level representations?

Method

Dataset: Subset of Human Connectome Project – Young Adult dataset

- Duration of scan: ~15 minutes
- Labels: Binary (good/bad label for each scan)
- Size of training set: 2536 scans, Validation set: 634 scans
- Training set contains only edge cases** (signals with uncertain labels are dropped)

Idea: While modelling physiological signals, dividing the signal into smaller segments helps us:

- Identify local artifacts → This entails informative representations.
- Use smaller model architectures → This allows faster convergence.

We approach the quality assessment problem in 2 ways:

- A Whole Signal Classification: Supervised Learning**
Determine if the full signal is good or bad (binary classification)

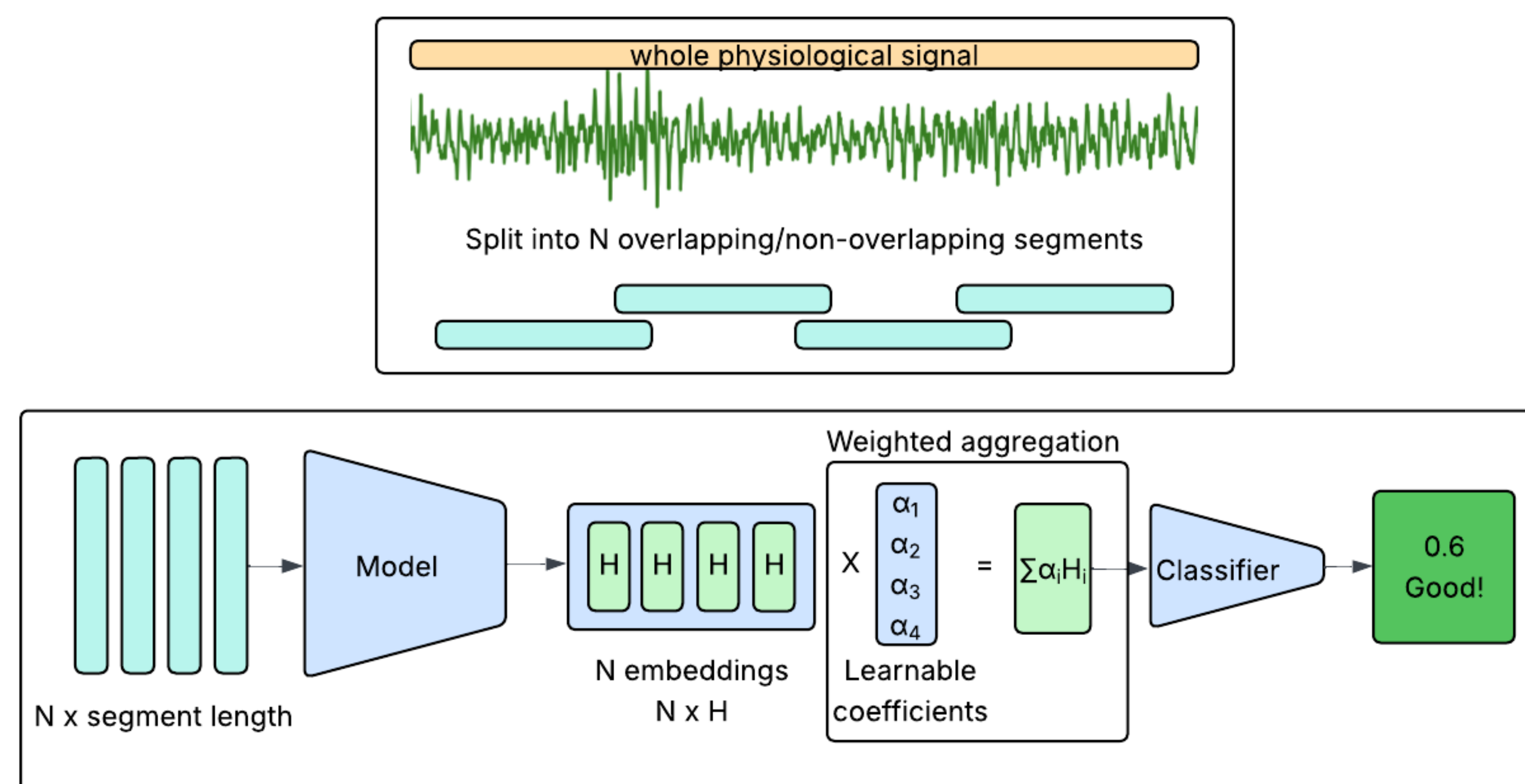


Figure 4: Framework for "segment-aggregation" i.e. obtaining embeddings per segment and aggregating them through a weighted linear combination prior to classification.

- B Error Detection: Semi-Supervised Learning**
Identifying specific anomalous segments in the signal

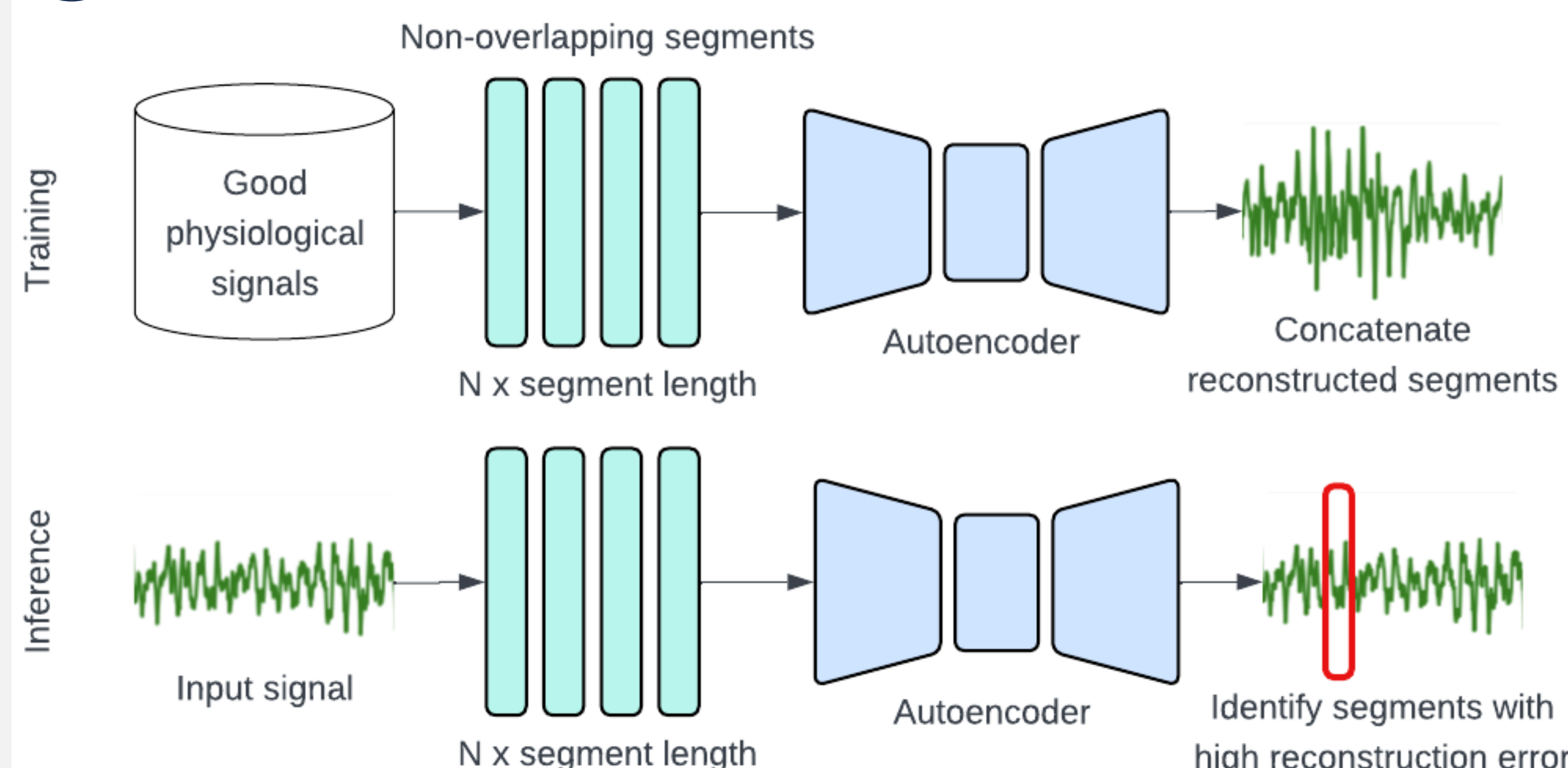


Figure 5: Framework for detection of anomalous regions by using only signals labelled as good in the training set

Results

A. Whole Signal Classification

We evaluate the segment-aggregation framework on a Bidirectional Long Short-Term Memory (BiLSTM) model as existing work shows its promise in physiological settings. [1]

Model Name	Segment Length	F1 Score	Accuracy
BiLSTM (full signal)	None	0.8638	0.8533
BiLSTM (w/ segments)	1024	0.8743	0.8754
BiLSTM (w/ segments)	512	0.8733	0.9038

Table 1: Segment-aggregation outperforms full signal classification

B. Error Detection

Segments with highest reconstruction errors ($> \text{mean} + 3 * \text{std}$) are highlighted as regions with potential anomalous artifacts.

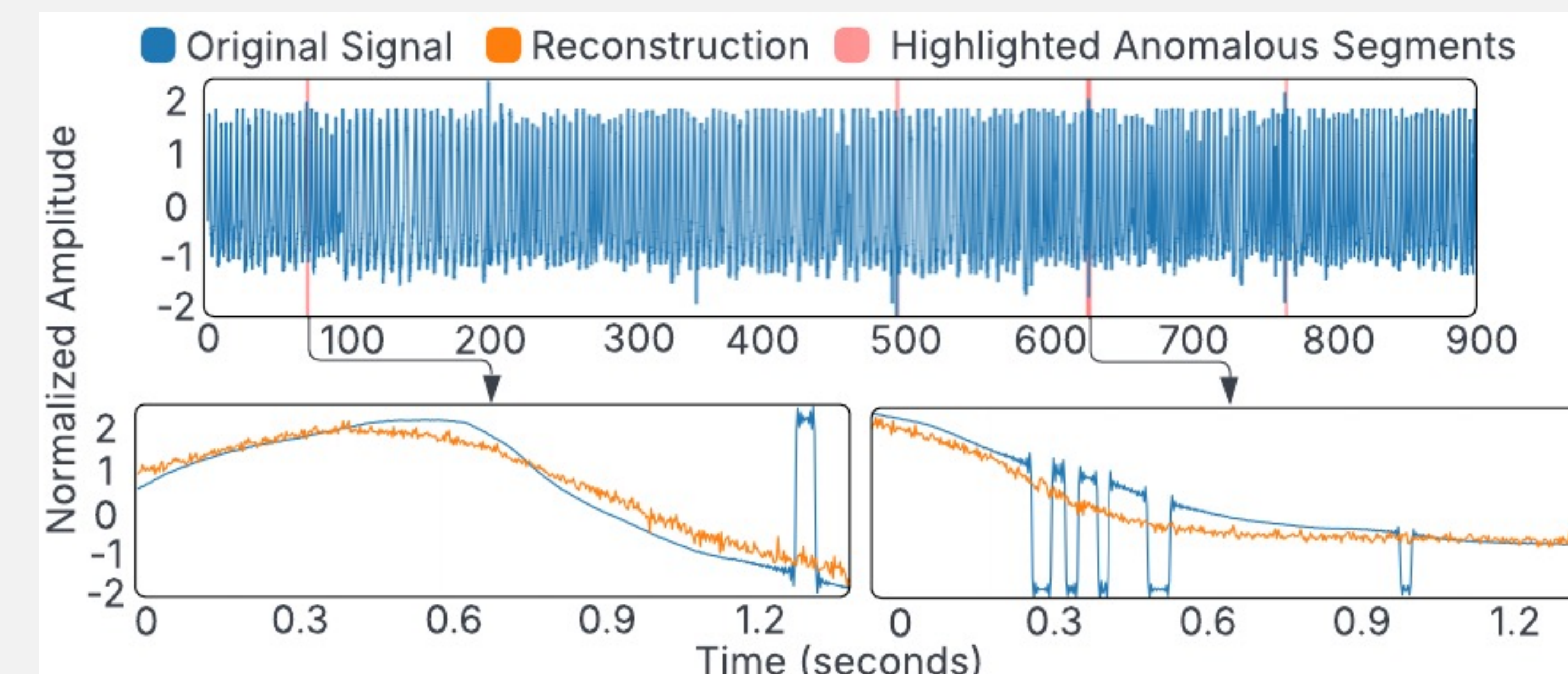


Figure 6: Zooming in on highlighted segments reveals anomalous spikes, which may be associated with motion artifacts during the scan.

Conclusion

Our deep learning framework, designed based on segment-level analysis of physiological signals, automates the quality assessment process with ~90% accuracy and reveals localized anomalous artifacts.

Future Directions

- Active learning offers a human-in-the-loop framework where the most informative samples are selected for manual labelling.
- Regressing over a quality metric instead of binary good/bad classes can be more meaningful for downstream applications.

Contributions & References

Author Contribution Statement: SS: Designed and implemented experiments, created all figures, tables and poster / RG, CC: Oversaw the project execution

[1] Bayrak, Roza G et al. "DeepPhysioRecon: Tracing peripheral physiology in low frequency fMRI dynamics." Imaging neuroscience (Cambridge, Mass.) vol. 3 IMAG.a.163. 25 Sep. 2025, doi:10.1162/IMAG.a.163